

Business Problem Statement

A retail company aims to gain a deeper understanding of customer purchasing behavior to improve business performance, enhance customer satisfaction, and strengthen long-term customer loyalty. The organization has observed variations in buying patterns across different age groups, genders, product categories, shopping modes (online and offline), and seasonal periods. Management wants to identify how factors such as discounts, customer reviews, payment methods, and purchase frequency influence customer decisions and overall sales.

The primary objective of this project is to analyze the customer shopping dataset and answer the following business question:

"How can customer shopping data be used to identify purchasing trends, improve customer engagement, and support better marketing and product decisions?"

Project Deliverables

1. Data Preparation and Transformation (Python)

Import the raw dataset, perform data cleaning, handle missing values, engineer new features, and prepare the data for analysis.

2. Database Analysis (MySQL)

Load the processed dataset into a MySQL database and execute SQL queries to analyze customer segments, purchasing behavior, loyalty patterns, revenue distribution, and other business metrics.

3. Interactive Dashboard (Power BI)

Design a dynamic Power BI dashboard with KPIs, charts, and slicers to visualize customer trends, sales performance, and key business insights for stakeholders.

4. Project Report and Presentation

Prepare a comprehensive project report documenting the complete analytics workflow, major findings, and business recommendations. Develop a presentation that effectively communicates insights to business stakeholders.

5. GitHub Repository

Maintain a well-organized GitHub repository containing the Python notebooks, MySQL scripts, Power BI dashboard file, project report, and supporting documentation to ensure reproducibility and easy collaboration.